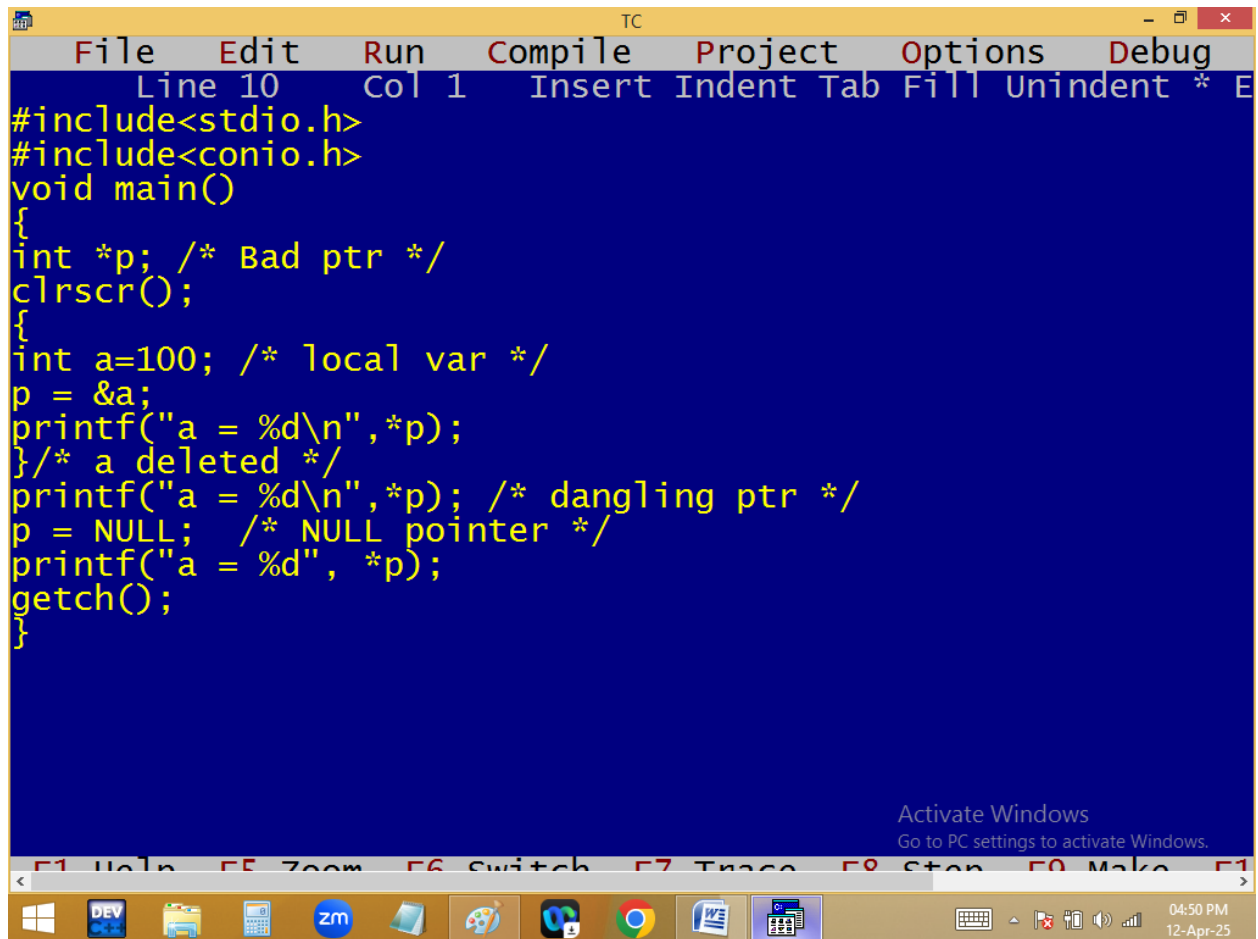


Dangling pointer:

A pointer is declared and some variable address also assigned. After some time that variable is deleted from memory. But the pointer is still pointing the deleted variable address. In this situation the pointer is called dangling pointer and to avoid this initialize the pointer with 0 or NULL.

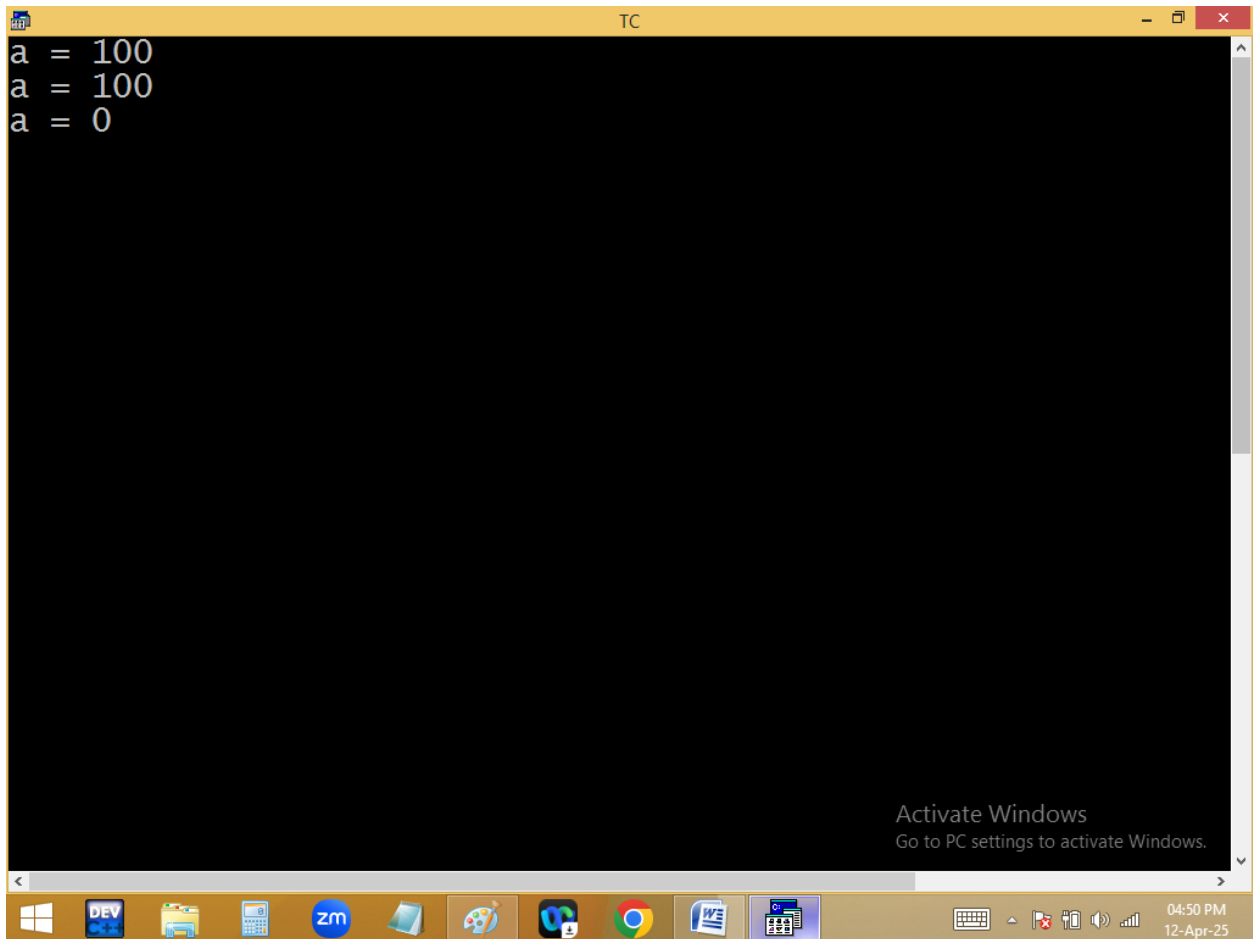


```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* Bad ptr */
clrscr();
{
int a=100; /* local var */
p = &a;
printf("a = %d\n",*p);
}/* a deleted */
printf("a = %d\n",*p); /* dangling ptr */
p = NULL; /* NULL pointer */
printf("a = %d", *p);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:50 PM
12-Apr-25



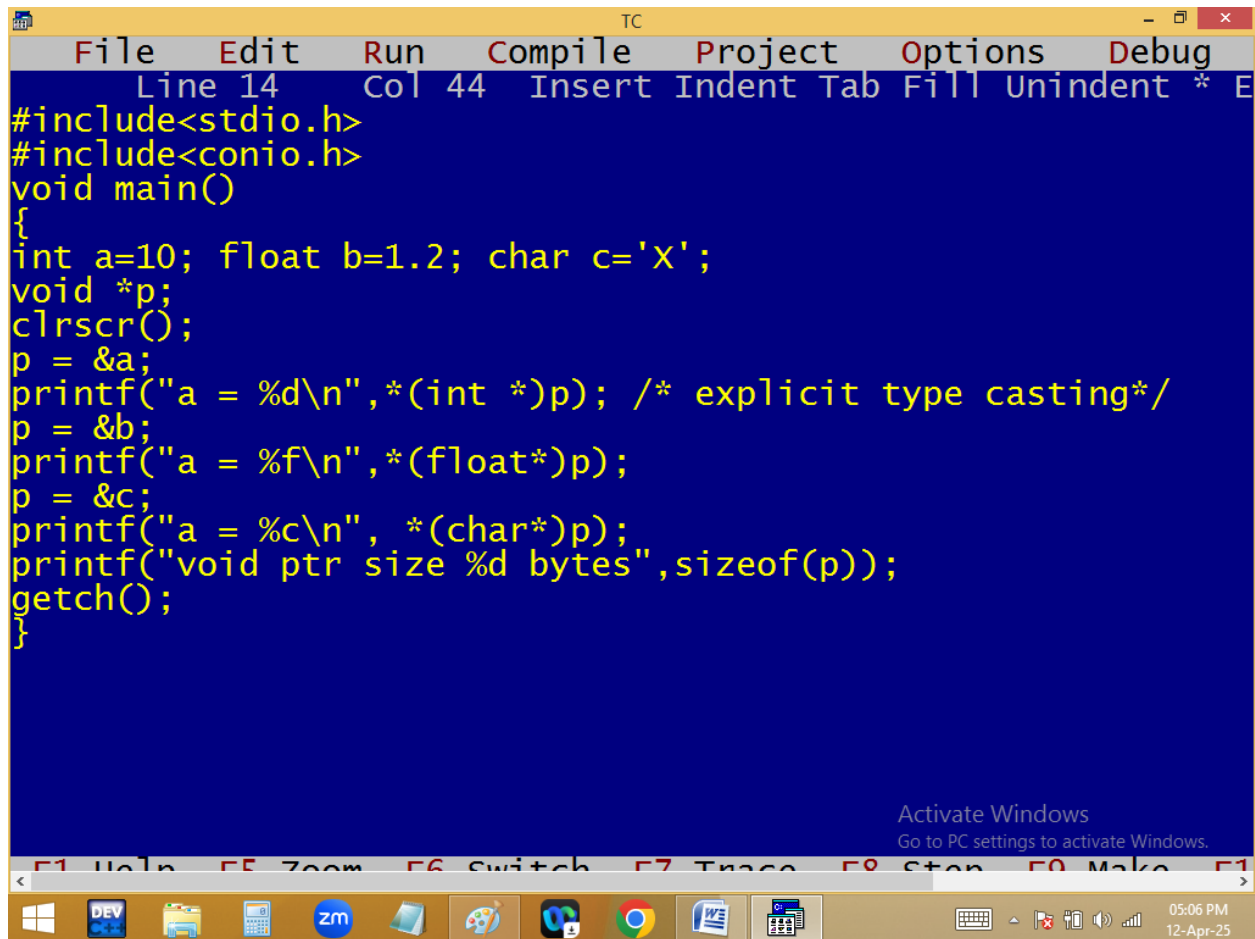
```
a = 100
a = 100
a = 0
```

void / generic pointer: pointer can store only the same type of variable address. When several variables with different data type, we have to declare several pointers.

Void pointer can store any type of variable address. Before going access void pointer, explicit type casting should be provided.

void pointer takes 2 bytes.

Void pointer very much used in dynamic memory allocation.

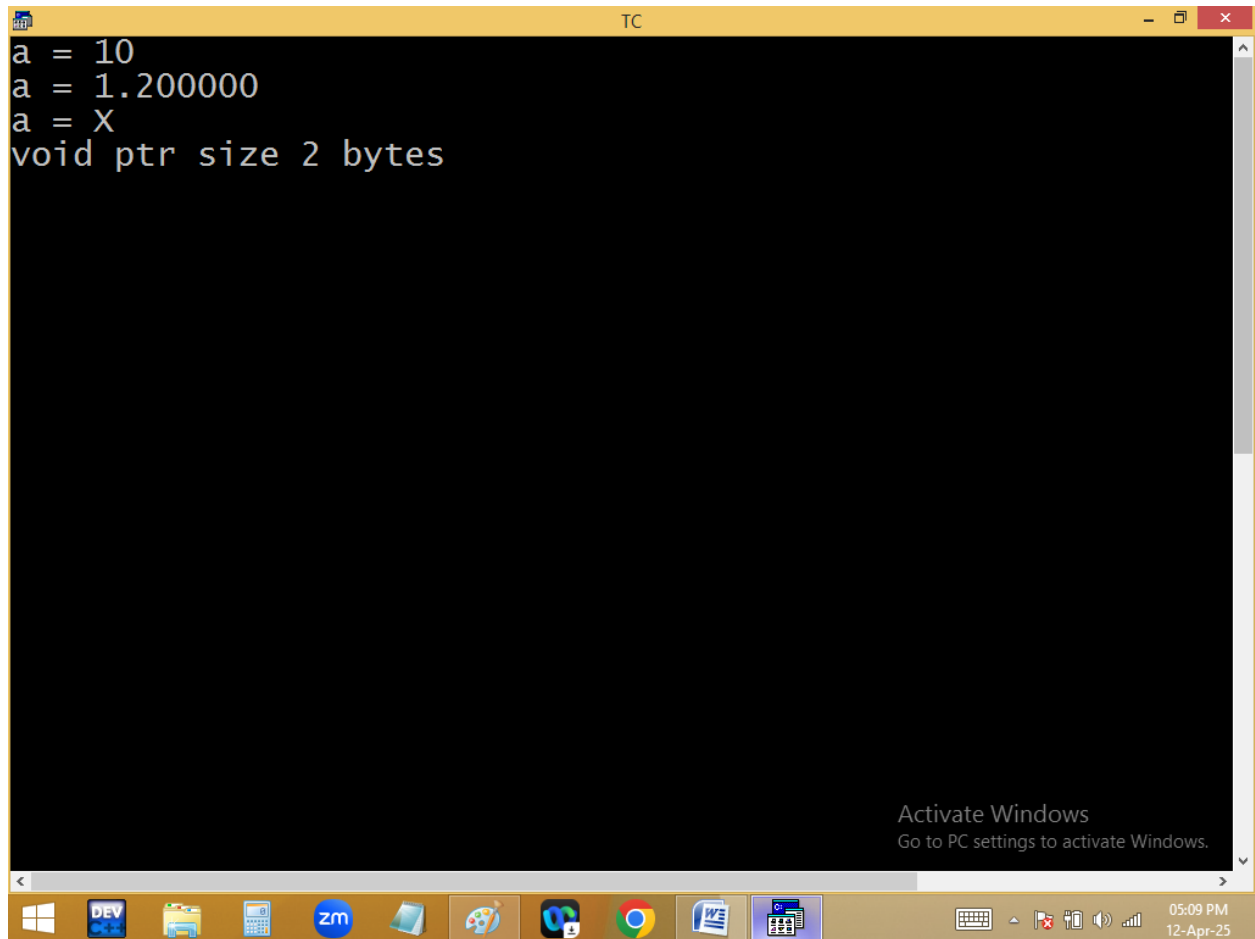


```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 44 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; float b=1.2; char c='X';
void *p;
clrscr();
p = &a;
printf("a = %d\n",*(int *)p); /* explicit type casting*/
p = &b;
printf("a = %f\n",*(float*)p);
p = &c;
printf("a = %c\n", *(char*)p);
printf("void ptr size %d bytes",sizeof(p));
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:06 PM
12-Apr-25



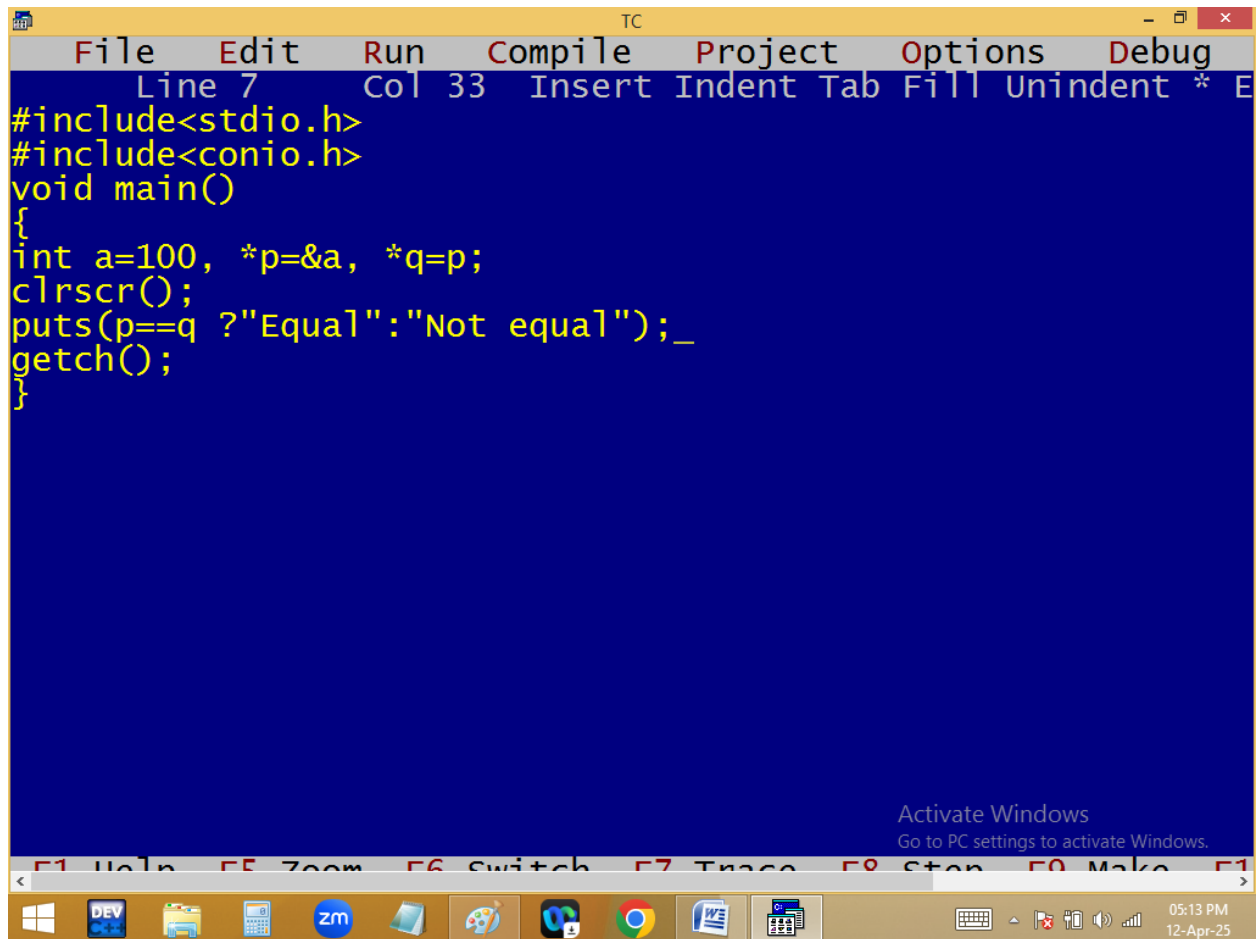
The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard Windows window controls (minimize, maximize, close). The main area is black with white text. The text displayed is:

```
a = 10  
a = 1.200000  
a = X  
void ptr size 2 bytes
```

At the bottom of the window, there is a taskbar with various icons including Windows, DEV, a folder, a calculator, a Zm icon, a paint palette, a code editor, a Chrome browser, a Word document, and a calculator. The system tray on the right shows the time as 05:09 PM and the date as 12-Apr-25. An "Activate Windows" watermark is visible in the bottom right corner of the window.

Pointer arithmetic:

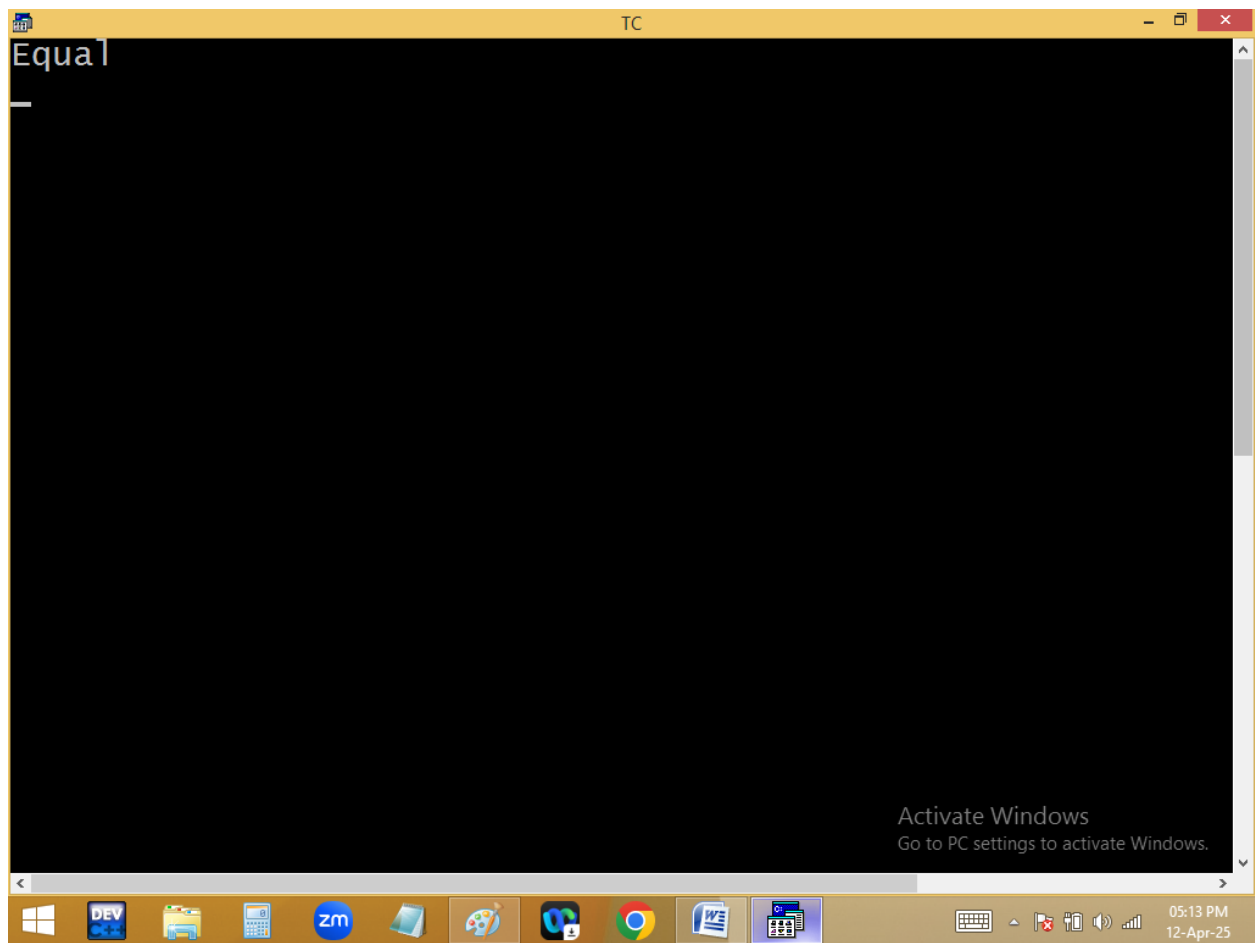
Like normal variables we can do `+`, `-`, `++`, `--`, `=`, `==` on pointers also. But we can't do `*`, `%`, `/` on pointers.

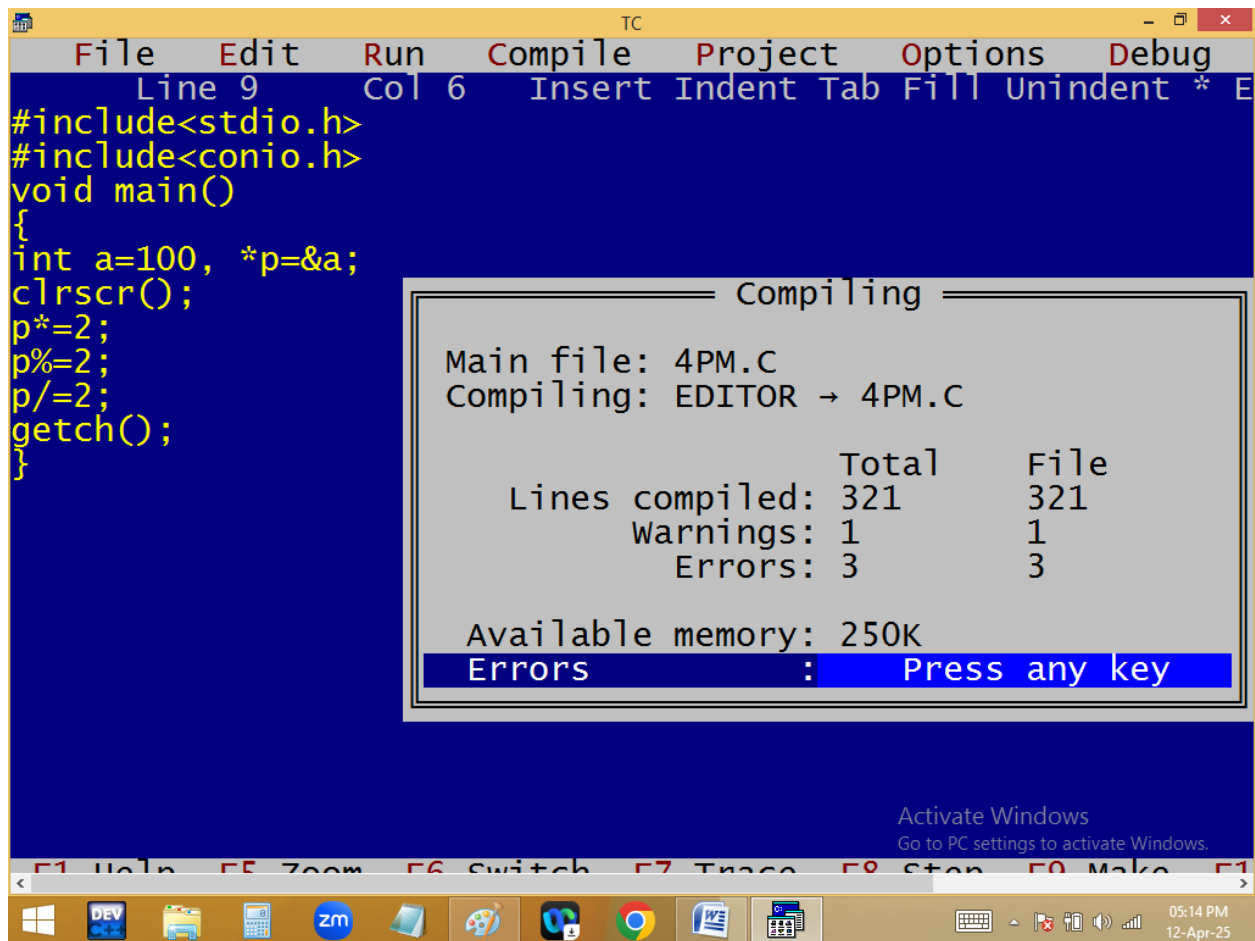


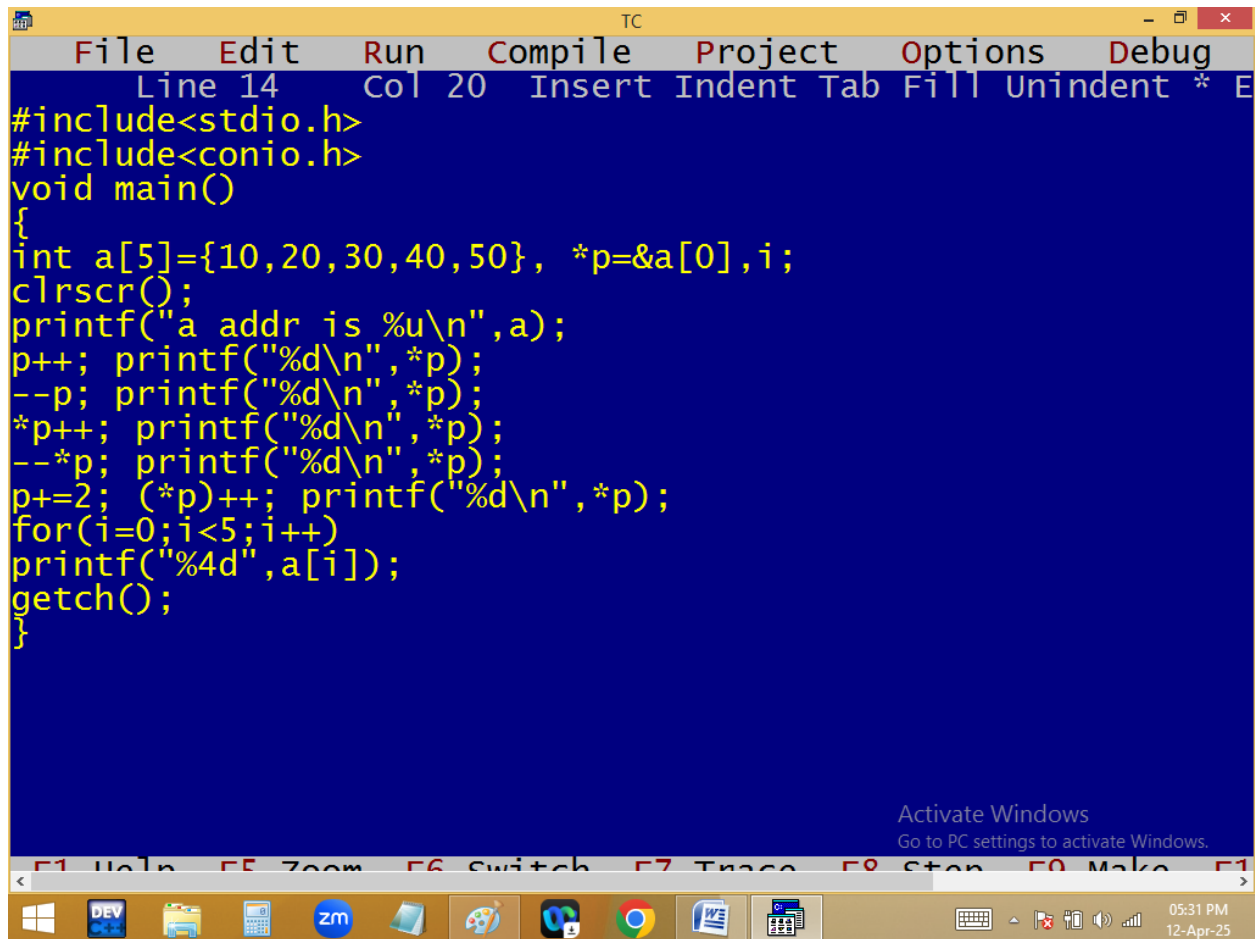
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 7" and "Col 33". The code editor has a blue background and contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a, *q=p;
clrscr();
puts(p==q ? "Equal": "Not equal");
getch();
}
```

At the bottom of the IDE, there is a toolbar with icons for various functions, and a status bar showing the time "05:13 PM" and date "12-Apr-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p=&a[0],i;
clrscr();
printf("a addr is %u\n",a);
p++; printf("%d\n",*p);
--p; printf("%d\n",*p);
*p++; printf("%d\n",*p);
--*p; printf("%d\n",*p);
p+=2; (*p)++; printf("%d\n",*p);
for(i=0;i<5;i++)
printf("%4d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

```

TC
a addr is 65494
20
10
20
19
41
10 19 30 41 50

```

Activate Windows
Go to PC settings to activate Windows.

p++; 65494+1*2==>65496

***p ==> value at 65496 ==> 20**

--p; ==>65496-1*2==>65494

***p ==> value at 65494 ==> 10**

***p++==>p++==> 65494+1*2=65496**

***p ==> value at 65496==>20**

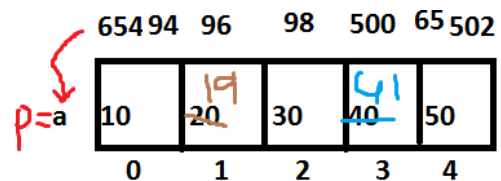
--*p ==> -- of value at 65496 ==>--20 ==> 19

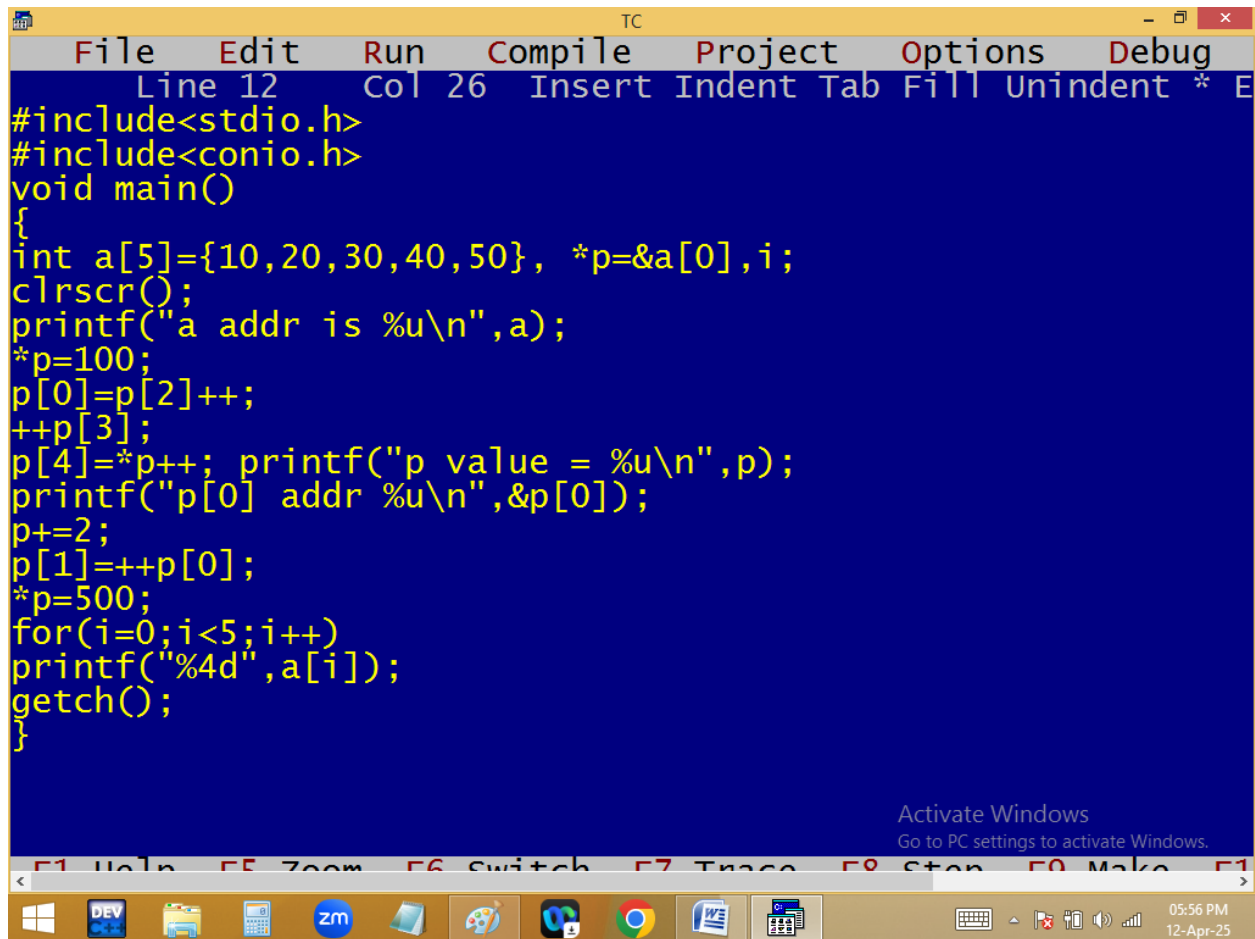
***p ==>value at 65496 ==> 19**

p+=2 ==>65496+2*2=65500

(*p)++ ==> value at 65500++ ==> 40++ ==> 41

***p ==> value at 65500 ==> 41**





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 12", "Col 26", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background with yellow text containing the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p=&a[0],i;
clrscr();
printf("a addr is %u\n",a);
*p=100;
p[0]=p[2]++;
++p[3];
p[4]=*p++; printf("p value = %u\n",p);
printf("p[0] addr %u\n",&p[0]);
p+=2;
p[1]=++p[0];
*p=500;
for(i=0;i<5;i++)
printf("%4d",a[i]);
getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Google Chrome, Word, and Excel. The system tray on the right shows the date and time: "05:56 PM 12-Apr-25".

